

Statistics

11

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In the 1880s Francis Galton kept a notebook of heights. He measured 928 grown children and the 205 sets of parents who raised them. He wanted to answer one question: how well does a parent's height predict a child's?

Tall parents did tend to have tall children. But Galton noticed something stranger. The children of very tall parents were, on average, a little shorter than their parents. The children of very short parents were a little taller. Each generation drifted back toward the middle. He called this "regression toward mediocrity", and although the insult has since been dropped, we still call the technique *regression* today.

To see the pattern at all, Galton first had to do something far more ordinary. He had 928 numbers. One number on its own says very little, and a thousand numbers say too much at once. He had to summarise them, measure how much they varied, and then compare one list against another.

That is what this chapter builds. You will learn to collect data honestly, and to describe a single list of numbers: where its centre lies and how widely it spreads. You will display a distribution as a histogram, a cumulative frequency curve or a box plot. Then you will compare two lists, measure whether they move together, and find the line that best predicts one from the other.

Every one of these tools has a range where it can be trusted and a range where it cannot, and knowing the difference is part of the subject. A correlation does not establish a cause. A line fitted between two values says nothing reliable beyond them. Galton's drift is worth holding on to as an example: it is not a fact about families, and no biology is needed to explain it. You will meet the reason at the end of the chapter.

11.1 Sampling and Data

You almost never have all the data. Galton could not measure every family in England, a factory cannot test every light bulb it makes, and a biologist cannot weigh every fish in a lake. So you measure a part and reason about the whole. The honesty of that reasoning depends entirely on how the part was chosen.

The complete group you want to know about is the **population**. The smaller group you actually measure is the **sample**. A number describing the population is a **parameter**; the matching number computed from a sample is a **statistic**. You use the statistic to estimate the parameter you cannot see.

DEF A **population** is the entire set of individuals under study. A **sample** is a subset of the population from which data are actually collected.

A **parameter** describes a population; a **statistic** describes a sample.

Types of data

Data come in two broad kinds. **Qualitative** data describe a quality or category, such as eye colour or country of birth. **Quantitative** data are numerical. Quantitative data split again. **Discrete** data take separate values you can count, such as the number of goals in a match. **Continuous** data can take any value in a range, such as height or time, limited only by the precision of your instrument.

The distinction matters because it decides how you may display and summarise the data. You can find the mean number of goals, but “mean eye colour” is meaningless.

Sampling techniques

A good sample resembles the population on a smaller scale. The techniques below are different ways of achieving that. Each one balances fairness against the effort required.

DEF **Simple random:** every member of the population has an equal chance of selection.

Systematic: order the population and select every k -th member from a random start.

Stratified: divide the population into groups (strata), then sample from each group in proportion to its size.

Quota: like stratified, but members within each group are chosen non-randomly until each quota is filled.

Convenience: select whoever is easiest to reach.

Stratified sampling is the most useful technique for populations with clear subgroups. If a school is 60% junior and 40% senior students, a stratified sample of 50 takes 30 juniors and 20 seniors, preserving the structure.

EXAMPLE 11.1

A school has 800 juniors and 400 seniors. A stratified sample of 60 students is taken. How many should be seniors?

Seniors are $\frac{400}{1200} = \frac{1}{3}$ of the school. The sample must keep that proportion:

$$\frac{1}{3} \times 60 = 20 \text{ seniors (and 40 juniors).}$$

Stratified sampling guarantees both groups are represented in their true ratio, which a simple random sample only achieves on average.

Bias and reliability

A sample is **biased** when its method systematically favours some outcomes over others. The statistic then misses the parameter, no matter how large the sample grows. A survey about reading habits conducted in a library will overstate how much people read. Bias is a fault in the method, not in the sample size; collecting more biased data only makes you more confident in the wrong answer.

Reliability also depends on the records themselves. Real data sets contain missing entries and recording errors. Before any analysis, ask how the data were collected, what might be missing, and whether any value could have been recorded wrongly.

CAUTION *A large sample is not automatically a good one. The 1936 Literary Digest poll surveyed 2.4 million people and still predicted the wrong US election winner. Its list of names came from telephone and car owners, who were wealthier than the average voter. Size cannot repair a biased method.*

Outliers

An **outlier** is a value far from the rest of the data. It may be a genuine extreme value, or it may be a recording error. Either way it can distort a summary badly. We will give it a pre-

cise definition once we have quartiles, in §11.4. For now, treat any suspiciously distant value as something to investigate, not something to delete immediately.

YOUR TURN 11.1 Sampling and Data

1. State whether each variable is discrete or continuous.

- (a) the number of goals scored in a match (b) the mass of a parcel
- (c) the number of students in a class (d) the time taken to run 100 m

2. A college has 1500 students, 600 of them male. A stratified sample of 50 is taken.

- (a) Find the number of males in the sample.
- (b) Find the number of females in the sample.
- (c) Explain one advantage of stratified sampling over simple random sampling here.

3. Sampling methods.

- (a) A sample of 40 is taken from 2000 people by choosing every k th person. Find k .
- (b) Name the sampling method in part (a).
- (c) A researcher surveys people at a gym about how much they exercise. Explain why this is biased.
- (d) A town of 15 000 has 4000 residents over 65. In a stratified sample of 150, find the number over 65.

11.2 Measures of Central Tendency

Once you have a list of numbers, the first question is always the same: what is a typical value? A single number that stands in for the whole list is a measure of **central tendency**. There are three, and they answer slightly different questions.

The **mean** is the balance point: add everything and share it equally. The **median** is the middle value when the data are ordered. The **mode** is the most frequent value.

The mean

For n raw values, the mean is the total divided by the count. When values repeat, you weight each value by how often it occurs, which is just a shortcut for the same sum.

KEY · IB

$$\bar{x} = \frac{\sum_{i=1}^k f_i x_i}{n}, \quad \text{where } n = \sum_{i=1}^k f_i$$

Here x_i are the distinct values, f_i their frequencies. A table listing each value alongside its frequency is called a **frequency distribution**, and the next example shows one. It is the standard compact way to present discrete data. With no repeats, every $f_i = 1$ and this collapses to $\bar{x} = \frac{\sum x_i}{n}$, the familiar form.

EXAMPLE 11.2

In 20 football matches a team scored the following numbers of goals. Find the mean.

Goals x	0	1	2	3	4
Frequency f	4	7	5	3	1

Multiply each value by its frequency, then divide by the total count:

$$\bar{x} = \frac{(0)(4) + (1)(7) + (2)(5) + (3)(3) + (4)(1)}{4 + 7 + 5 + 3 + 1} = \frac{30}{20} = 1.5.$$

The mean need not be a value the data can actually take. No team scores 1.5 goals, but 1.5 is the fair share.

Median and mode

The median splits ordered data into two equal halves. For n values, it sits at position $\frac{n+1}{2}$. If n is odd this is a single middle value; if n is even it is the average of the two middle values.

EXAMPLE 11.3

Find the median and mode of 4, 7, 2, 9, 4, 11, 6.

Order first: 2, 4, 4, 6, 7, 9, 11. With $n = 7$, the median is at position $\frac{7+1}{2} = 4$, the fourth value: 6.

The mode is the most frequent value: 4 appears twice, everything else once, so the mode is 4.

Grouped data

Continuous data are often recorded in classes, such as “time between 10 and 20 minutes.” You no longer know the individual values, only the class each fell into. To estimate the mean, assume every value sits at its class **midpoint**, then apply the frequency formula. The result is an estimate, because the assumption is rarely exact.

EXAMPLE 11.4

Fifty phone calls to a help line had the durations below. Estimate the mean duration.

Duration (min)	$0 \leq t < 4$	$4 \leq t < 8$	$8 \leq t < 12$	$12 \leq t < 16$
Frequency	8	18	16	8

Use the midpoint of each class: 2, 6, 10, 14.

$$\bar{x} = \frac{(2)(8) + (6)(18) + (10)(16) + (14)(8)}{50} = \frac{16 + 108 + 160 + 112}{50} = \frac{396}{50} = 7.92 \text{ min.}$$

The **modal class** is $4 \leq t < 8$, the class with the highest frequency. With grouped data you report a modal class, not a single modal value.

Which measure to use

The three measures disagree when data are skewed, and that disagreement tells you something. The mean uses every value, so a single very large value raises it. The median depends only on position, so it changes very little. This is why incomes are reported by median, not mean: a few billionaires would make the mean meaningless as a “typical” wage.

CAUTION *The mean is sensitive to outliers; the median is resistant to them. When a distribution is strongly skewed or contains a suspect extreme value, the median usually describes the centre better.*

YOUR TURN 11.2 Measures of Central Tendency

4. For the data 5, 8, 8, 10, 14, 15, find each of the following.

- | | |
|--------------|----------------|
| (a) the mean | (b) the median |
| (c) the mode | (d) the range |

5. Find the required measure in each case.

- | | |
|--|--|
| (a) the median of 3, 7, 9, 12, 15, 20 | (b) the mode of 2, 4, 4, 5, 7, 7, 7, 9 |
| (c) the mean of 3, 7, 7, 8, 10, 11, 12, 14 | (d) the median of 12, 15, 15, 18, 20, 22 |

6. Work backwards from a given mean.

- | |
|--|
| (a) The numbers 4, 7, x , 12, 15 have a mean of 10. Find x . |
| (b) The mean of six numbers is 14. A seventh number is added and the mean becomes 15. Find the seventh number. |

7. The table shows a frequency distribution.

Value x	1	2	3	4	5
Frequency f	3	5	8	4	2

- (a) Find n .
 (b) Find $\sum fx$.
 (c) Find the mean.
 (d) State the mode.

8. The grouped data below record a measurement x .

Class	$0 \leq x < 10$	$10 \leq x < 20$	$20 \leq x < 30$	$30 \leq x < 40$
Frequency	6	10	9	5

- (a) Write down the four class midpoints.
 (b) Estimate the mean.
 (c) State the modal class.
 (d) Explain why the mean is only an estimate.

11.3 Measures of Dispersion

The centre alone does not describe a data set. Consider two classes that both average 60% on a test. In the first, every student scored between 58 and 62. In the second, half scored 30 and half scored 90. The mean is the same; the two classes are not. To capture the difference you need a measure of **spread**.

Range and interquartile range

The simplest measure is the **range**: the largest value minus the smallest. It is easy to find but unreliable. It depends only on the two most extreme values, which are exactly the ones most likely to be outliers.

A better approach is to ignore the extremes and measure the spread of the middle. Just as the median cuts the data in half, the **quartiles** cut it into quarters. Q_1 is the value one quarter of the way through the ordered data, Q_2 is the median, and Q_3 is three quarters of the way. The spread of the central half is the **interquartile range**.

KEY · IB

$$\text{IQR} = Q_3 - Q_1$$

Because it discards the top and bottom quarters, the IQR is unaffected by outliers, which is why it is normally reported alongside the median.

EXAMPLE 11.5

Find the range and interquartile range of

$$3, 5, 6, 8, 8, 9, 11, 13, 15.$$

The data are already ordered, with $n = 9$. The range is $15 - 3 = 12$.

The median (Q_2) is the 5th value, 8. The lower quartile is the median of the four values below it, 3, 5, 6, 8:

$$Q_1 = \frac{5 + 6}{2} = 5.5.$$

The upper quartile is the median of the four values above, 9, 11, 13, 15:

$$Q_3 = \frac{11 + 13}{2} = 12.$$

So $IQR = 12 - 5.5 = 6.5$. Methods for placing quartiles vary slightly between textbooks and calculators; in this course, report the value your GDC gives.

Variance and standard deviation

The IQR uses only the middle half of the data. A measure that uses every value would be better. The idea: measure how far each value sits from the mean, then average those distances.

You cannot simply average the deviations $x_i - \bar{x}$, because by the definition of the mean they cancel to zero. The solution is to square them first, which removes the signs. The average squared deviation is the **variance**; its square root, which returns to the original units, is the **standard deviation**.

KEY

$$\sigma^2 = \frac{\sum_{i=1}^k f_i(x_i - \bar{x})^2}{n}, \quad \sigma = \sqrt{\sigma^2}$$

The standard deviation σ is, roughly, the typical distance of a value from the mean. A small σ means the values lie close to the centre. A large σ means they are widely spread.

CAUTION In this course you find σ with your **GDC** (graphic display calculator), not by hand. The formula is given so that you understand what the calculator computes, not so that you evaluate it by hand under exam conditions. Know what σ means.

EXAMPLE 11.6

Return to the two classes, both with mean 60. Class A scored 58, 59, 60, 61, 62; Class B scored 30, 45, 60, 75, 90. Compare their standard deviations.

Both means are 60. For Class A the deviations are $-2, -1, 0, 1, 2$:

$$\sigma_A = \sqrt{\frac{4 + 1 + 0 + 1 + 4}{5}} = \sqrt{2} \approx 1.41.$$

For Class B the deviations are $-30, -15, 0, 15, 30$:

$$\sigma_B = \sqrt{\frac{900 + 225 + 0 + 225 + 900}{5}} = \sqrt{450} \approx 21.2.$$

The means are identical, but the standard deviations differ by a factor of 15. The single number σ captures exactly the difference that the mean alone does not show.

The effect of changing the data

Suppose every value in a data set is changed in the same way, either by adding a constant or by multiplying by a constant. Temperatures convert from Celsius to Fahrenheit; prices rise by a fixed tax. You can predict the new mean and standard deviation without recomputing anything.

Adding a constant c to every value shifts the whole distribution sideways. The centre moves by c , but the spread does not change. Multiplying every value by k stretches the distribution away from zero, scaling both the centre and the spread.

KEY If $y_i = kx_i + c$ for every value, then

$$\bar{y} = k\bar{x} + c, \quad \sigma_y = |k| \sigma_x, \quad \sigma_y^2 = k^2 \sigma_x^2.$$

The constant c does not affect the spread. Moving every value by the same amount changes no distance between them. Only the multiplier k rescales σ .

EXAMPLE 11.7

A set of test marks has mean 54 and standard deviation 8. Every mark is scaled by multiplying by 1.5 and then adding 5. Find the new mean and standard deviation.

Here $k = 1.5$ and $c = 5$:

$$\bar{y} = 1.5(54) + 5 = 86, \quad \sigma_y = 1.5 \times 8 = 12.$$

The $+5$ moves the mean but does not change the spread. Only the $\times 1.5$ affects σ .

YOUR TURN 11.3 Measures of Dispersion

9. For the data 2, 4, 5, 7, 8, 10, 12, 14, 16, 20, find each of the following.

- (a) the range
- (b) Q_1
- (c) Q_3
- (d) the interquartile range

10. GDC Use your GDC for each part.

- (a) Find the mean and standard deviation of 5, 8, 9, 11, 13, 14.
- (b) Find the mean and standard deviation of the values 10, 20, 30, 40 with frequencies 2, 5, 8, 3.
- (c) Find the standard deviation of 12, 15, 15, 18, 20, 22.

11. Each data set is transformed. Find the new mean and the new standard deviation.

- (a) Mean 50, standard deviation 6; every value is multiplied by 3.
- (b) Mean 30, standard deviation 5; the value 10 is added to every data point.
- (c) Mean 60, standard deviation 12; each value is transformed by $y = 0.5x + 20$.
- (d) Mean 40, standard deviation 9; each value is transformed by $y = 2x - 5$.

12. Explain each of the following.

- (a) Why adding a constant to every value leaves the standard deviation unchanged.
- (b) Why the interquartile range is preferred to the range when the data may contain outliers.

11.4 Visualising Distributions

Numbers summarise a data set; a picture shows its shape. A mean and standard deviation tell you where the data sit and how far they spread. They do not tell you whether the distribution is symmetric, or leans to one side, or is concentrated at one end. Three kinds of display are used in this course.

| Histograms

A **histogram** groups continuous data into classes and draws a bar for each, with height equal to the frequency. Unlike a bar chart for categories, the bars touch, because the horizontal axis is a continuous number line. In this course every class has the same width.

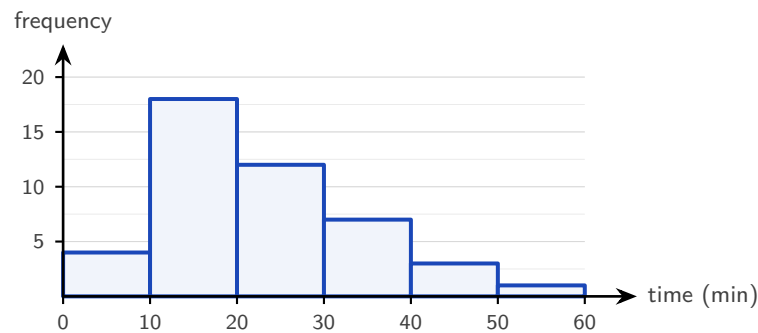


Figure 11.1 A histogram of 45 waiting times in classes of equal width. The bars touch because the horizontal axis is a continuous number line. The long tail to the right is positive skew.

The shape is what a histogram is for. It may be a symmetric peak, a long tail to the right (**positive skew**), as above, or a long tail to the left (**negative skew**).

In a positively skewed distribution the few large values raise the mean above the median. In a negatively skewed one the order reverses. Comparing mean and median is a quick numerical test for skew.

Cumulative frequency

A **cumulative frequency** curve answers “how many values fall below this point?” You plot the running total of frequencies against the *upper* boundary of each class, then join the points with a smooth curve. The result is an S-shape, and the median and quartiles can be read from it directly. Enter from the vertical axis at one quarter, one half and three quarters of the total. Move across to the curve, then drop to the horizontal axis to read Q_1 , Q_2 and Q_3 .

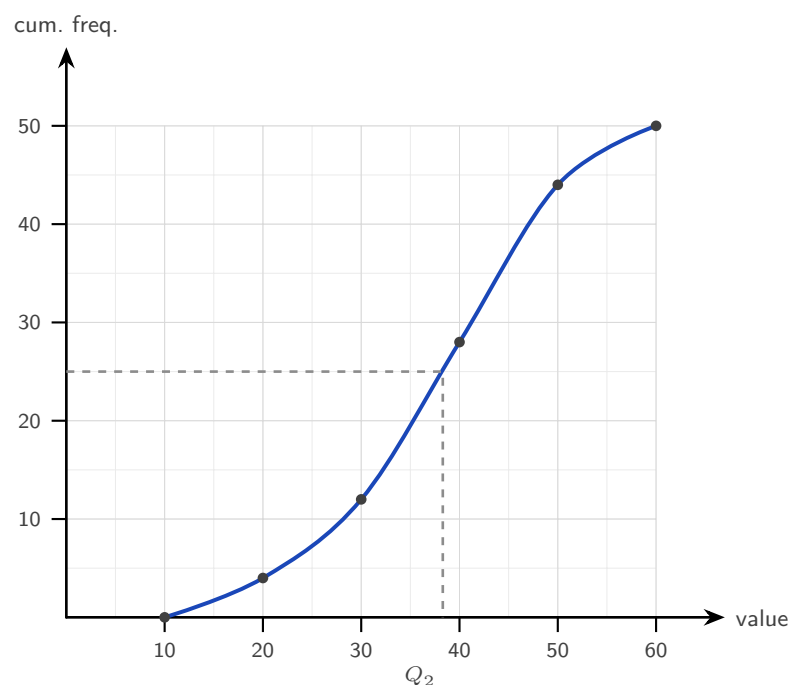


Figure 11.2 A cumulative frequency curve for 50 values. Each point is plotted at the *upper* boundary of its class. The dashed line reads the median: enter at half the total frequency, cross to the curve, drop to the value axis.

The dashed line shows the median. Enter at half the total frequency, here 25 of 50, cross to the curve, then drop to the value axis. Note that each point is plotted at the *upper* boundary of its class, and the curve is drawn through the points.

The same read-off gives any **percentile**: the p -th percentile is the value below which $p\%$ of the data lie. Enter the vertical axis at $\frac{p}{100} \times n$ and read across and down exactly as for the median. The median is the 50th percentile, and the quartiles Q_1 and Q_3 are the 25th and 75th. For the curve above, the 90th percentile is read off at a cumulative frequency of $0.9 \times 50 = 45$, giving a value of roughly 51.

Box-and-whisker plots

A **box-and-whisker plot** draws the **five-number summary**: minimum, Q_1 , median, Q_3 , maximum. The box spans the interquartile range, and a line inside it marks the median. The whiskers reach to the smallest and largest values that are *not* outliers. Any outlier is marked separately with a cross. In one small picture you see the centre, the spread, and the skew.

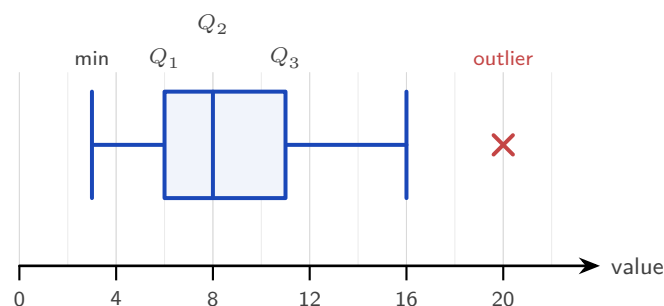


Figure 11.3 A box-and-whisker plot of the five-number summary. The box spans the interquartile range and the inner line marks the median. The right whisker stops at 16, the largest value that is not an outlier; the value 20 is marked with a cross.

A box positioned to the left, with a long right whisker, as above, indicates positive skew. Here the value 20 is an outlier, by the rule below. The whisker therefore stops at 16, the largest value that is not an outlier, and 20 is marked with a cross. Comparing two box plots side by side is the fastest way to contrast two distributions.

A box plot also gives a first hint about the normal distribution of Ch. 13. If the box and whiskers are roughly symmetric about the median, the data *may* be normally distributed. Clear skew shows that they are not.

The outlier rule

Now “outlier” can be defined precisely. A value is treated as an outlier if it lies more than 1.5 interquartile ranges beyond the nearer quartile.

KEY A value x is an **outlier** if

$$x < Q_1 - 1.5(\text{IQR}) \quad \text{or} \quad x > Q_3 + 1.5(\text{IQR}).$$

EXAMPLE 11.8

A data set has $Q_1 = 24$, $Q_3 = 40$. Is a value of 70 an outlier?

First $\text{IQR} = 40 - 24 = 16$. The upper boundary is

$$Q_3 + 1.5(\text{IQR}) = 40 + 1.5(16) = 40 + 24 = 64.$$

Since $70 > 64$, the value is an outlier. On a box plot it would be marked with a cross, with the whisker stopping at the largest value that is not an outlier.

YOUR TURN 11.4 Visualising Distributions

13. Consider the data 4, 6, 7, 9, 10, 12, 13, 15, 18.

- (a) Find the median. (b) Find Q_1 and Q_3 .
(c) Write down the five-number summary. (d) State the interquartile range.

14. Apply the $1.5 \times \text{IQR}$ outlier rule.

- (a) A data set has $Q_1 = 30$ and $Q_3 = 50$. Determine whether 95 is an outlier.
(b) A data set has $Q_1 = 12$ and $Q_3 = 20$. Determine whether 2 is an outlier.
(c) A data set has $Q_1 = 9$ and $Q_3 = 19$. Find both outlier boundaries.

15. Reading displays.

- (a) A box plot has $Q_1 = 12$ and $Q_3 = 24$. Find the interquartile range.
(b) A cumulative frequency curve is drawn for 80 values. State the cumulative frequencies at which you read off Q_1 , the median and Q_3 .
(c) Explain why the bars of a histogram touch, while those of a bar chart do not.

11.5 Correlation

Everything so far has described one variable. Galton's question was different: he had *two* measurements on each family, parent height and child height, and wanted to know whether

they were linked. This is the study of **bivariate** data, meaning data in which two variables are recorded for each individual. The first step is always to look.

Scatter diagrams

A **scatter diagram** plots each pair (x, y) as a point. The cloud of points has a shape, and the shape is the relationship. If y tends to rise as x rises, the correlation is **positive**. If y falls as x rises, it is **negative**. If there is no tilt, there is no linear correlation. The tighter the points cluster around a straight line, the **stronger** the correlation.

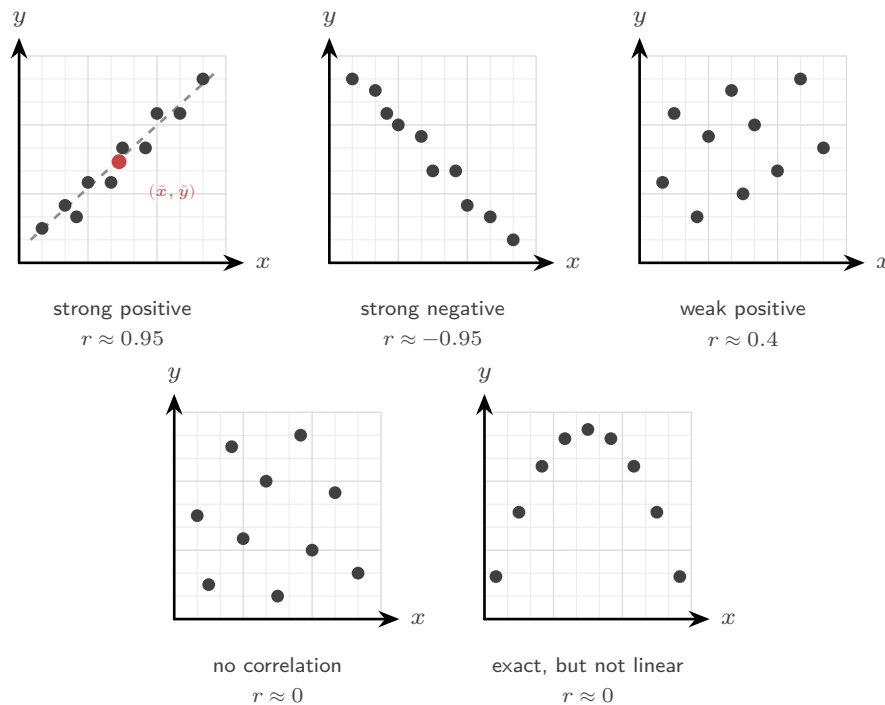


Figure 11.4 Five clouds and the value of r each gives. The sign of r is the direction of the tilt and the size of r is how tightly the points hug a line. The last panel is the warning: the points lie exactly on a curve, so the relationship could not be stronger, yet r is near zero because the curve is not straight.

Read the panels from left to right. The first two tilt clearly, one up and one down, and the points sit close to a line: those are strong correlations, positive and negative. The third tilts upward but the points scatter widely, so the correlation is weak. The fourth has no tilt at all.

The last panel is the one to remember. Its points lie exactly on a curve, so the two variables are perfectly related, yet r is near zero. That is because r only measures how close the points are to a *straight* line.

The first panel also shows a **line of best fit** drawn by eye: a straight line balancing the points above and below it. To position it, first compute the **mean point** $(\bar{x}, \bar{y})$, the mean of the x -values paired with the mean of the y -values, marked in scarlet. Draw your line through it. Every correctly drawn line of best fit passes through the mean point; in §11.6 the calculator makes this exact.

Pearson's correlation coefficient

You can judge correlation well by eye, but not precisely. **Pearson's product-moment correlation coefficient**, written r , turns the strength and direction into a single number between -1 and 1 .

DEF Pearson's correlation coefficient r measures the strength and direction of the **linear** relationship between two variables. It satisfies $-1 \leq r \leq 1$.

The sign of r gives the direction, and its size gives the strength. The extremes $r = \pm 1$ mean the points lie exactly on a straight line; $r = 0$ means no linear relationship. A rough reading:

$ r $ near	Interpretation
1	strong linear correlation
0.5	moderate linear correlation
0	weak or no linear correlation

You compute r with your GDC by entering the two lists. Reading and interpreting r matters far more than the formula behind it.

CAUTION r measures linear association only. Data lying perfectly on a parabola can have r near 0, because the relationship, though exact, is not straight. Always look at the scatter diagram before trusting r .

Correlation is not causation

A strong r says two variables move together. It does not say one causes the other. There are at least three quite different reasons a correlation can appear without any causal link, and it is worth being able to tell them apart.

A third variable drives both. Ice-cream sales and drowning deaths rise and fall together across the year. Neither causes the other. Summer heat raises both, because hot weather sends people to the ice-cream shop and to the water. A variable like this, sitting behind both and explaining the link, is called a **confounding variable**.

The arrow points the other way. Hospitals with the most advanced equipment can show the highest death rates. The equipment is not killing patients. The most seriously ill patients are sent to the best-equipped hospitals, so the severity of illness is causing the choice of hospital, not the reverse.

Coincidence, found by searching. Between 2000 and 2009, cheese consumption per person in the United States tracked the number of people who died by becoming entangled in their bedsheets, with r above 0.94. Nobody thinks these are connected. If you compare thousands of pairs of unrelated variables, some pairs will match closely by chance alone, and

a determined search will always find them.

The third case is the one to keep in mind whenever a surprising correlation is reported. A high r is easy to find if you go looking through enough data.

CAUTION *A high correlation never establishes cause. There may be a hidden third variable, or the link may be coincidence. Causation must be argued from the context, never from r alone.*

YOUR TURN 11.5 Correlation

16. Describe the correlation indicated by each value of r .

(a) $r = -0.85$

(b) $r = 0.12$

(c) $r = 0.97$

(d) $r = -0.40$

17. **GDC** Find Pearson's correlation coefficient for each data set.

(a) $x = 1, 2, 3, 4, 5$ and $y = 2, 5, 4, 7, 9$

(b) $x = 2, 4, 6, 8, 10$ and $y = 5, 9, 10, 14, 17$

18. Interpreting r .

(a) A set of points lies exactly on the parabola $y = x^2$ for $-3 \leq x \leq 3$. Explain why r is close to 0 even though the relationship is exact.

(b) A study finds a strong positive correlation between shoe size and reading ability in children. Explain why shoe size does not cause reading ability.

11.6 Linear Regression

If the scatter diagram shows a clear linear trend, you can do more than measure it. You can draw the line that best fits the data and use it to predict. The question is what “best” should mean.

The least-squares line

For any candidate line, each data point sits some vertical distance above or below it. That distance is the **residual**, the error of the line at that point. The **least-squares regression line** is the one line that makes the total of the squared residuals as small as possible. Squaring, again, stops positive and negative errors from cancelling, and it counts large errors much more heavily than small ones.

KEY The **regression line of y on x** has the form

$$y = ax + b,$$

chosen to minimise the sum of squared vertical residuals. It always passes through the mean point $(\bar{x}, \bar{y})$.

You find a and b from your GDC. You must also be able to say what they *mean* in context. The gradient a is the change in y for each one-unit increase in x . The intercept b is the predicted y when $x = 0$. The line always passes through $(\bar{x}, \bar{y})$, which is a useful check and an occasional exam shortcut.

EXAMPLE 11.9

A GDC gives the regression line of exam score y on hours studied x as $y = 6.2x + 41$, valid for students who studied between 2 and 9 hours. Predict the score of a student who studied 5 hours, interpret the values 6.2 and 41 in context, and comment on predicting the score for 20 hours.

For $x = 5$, substitute into the line:

$$y = 6.2(5) + 41 = 31 + 41 = 72.$$

In context, the gradient says each additional hour of study raises the predicted score by 6.2 marks. The intercept 41 is the predicted score of a student who did not study at all. That reading is itself a mild extrapolation, because $x = 0$ lies below the data range.

Predicting at $x = 20$ is unsafe. The data only cover 2 to 9 hours, so 20 lies far outside the known range, and the linear trend may not continue there.

Interpolation and extrapolation

Predicting *inside* the range of the data is **interpolation**, and it is reliable when the correlation is strong. Predicting *outside* the range is **extrapolation**, and it is unreliable. You are assuming the pattern continues where you have no evidence. The further beyond the data you go, the less trustworthy the prediction.

CAUTION A regression line is only trustworthy for prediction within the range of the original data, and only when $|r|$ is reasonably large. Extrapolating far beyond the data, or fitting a line to weakly correlated points, produces predictions worth little.

The two regression lines

There is an asymmetry hidden in “least squares.” The line of y on x minimises *vertical* residuals, because it treats y as the quantity to predict from x . If instead you want to predict x from y , you minimise *horizontal* residuals, giving a different line: the **regression line of x on y** .

KEY Use the regression line of y on x to predict y from a given x . Use the regression line of x on y to predict x from a given y . Both lines pass through $(\bar{x}, \bar{y})$, and they coincide only when $r = \pm 1$.

The rule is simple. Choose the line whose name begins with the quantity you want to predict. Choosing the wrong line is a common and avoidable error.

EXAMPLE 11.10

For a set of bivariate data, $\bar{x} = 10$ and $\bar{y} = 25$. The regression line of y on x is $y = 2x + 5$. Verify that the mean point lies on this line.

Substitute $x = \bar{x} = 10$:

$$y = 2(10) + 5 = 25 = \bar{y}. \checkmark$$

Every least-squares line of y on x passes through $(\bar{x}, \bar{y})$, so this is a quick check on a computed line.

YOUR TURN 11.6 Linear Regression

19. **GDC** Find the regression line of y on x for each data set.

- (a) $x = 1, 2, 3, 4, 5$ and $y = 3, 5, 4, 8, 10$
- (b) $x = 2, 4, 6, 8, 10$ and $y = 5, 9, 10, 14, 17$

20. Using a regression line.

- (a) A regression line is $y = 2.5x + 4$. Predict y when $x = 6$.
- (b) A regression line $y = -1.5x + 30$ was fitted using data with $2 \leq x \leq 15$. Predict y when $x = 8$.
- (c) Explain why using that same line to predict y at $x = 40$ is unreliable.

21. Two further results.

- (a) The mean point of a data set is $(8, 20)$ and the regression line of y on x has gradient 1.5. Find the equation of the line.
- (b) You are given a value of y and want to predict the corresponding x . State which regression line you should use, and why.

Chapter 11 · Statistics

Reflections

TOK Almost everything you are told about the world arrives as a number. Your grade. The mass of an atom. How happy a country is. How intelligent you are. Each is reported as a figure, and the figure is then treated as the thing itself.

But mass really is a quantity, and 12.011 is a fair description of carbon. Happiness is not a quantity in that sense. A score of 7.2 is the output of a questionnaire somebody designed, with questions somebody chose, and a grade of 6 compresses two years of work into one digit.

So every statistic makes two claims: that the thing measured has a size at all, and that this particular number captures it. Both can fail, and they fail independently. The arithmetic still works perfectly, and the conclusion can still be nonsense.

Which of the numbers used to describe you measure something that was already there, and which create the thing they claim to measure?

RW Galton's "regression toward mediocrity" is now called regression to the mean, and it affects decision-making everywhere. A pilot who flies badly and is scolded usually does better next time. The reason is not the scolding. An extreme performance is, on average, followed by a less extreme one. Mistaking this statistical drift for cause and effect has misled trainers, investors, and doctors for over a century.

The same trap sits in school data. A class that scores unusually low one term tends to score closer to average the next, whatever was changed in between, so any change made after a bad term will appear to have worked. It is also why a drug trial needs a control group: patients enrol when their illness is at its worst, and an illness measured at its worst tends to be measured better next time on its own.

IM The word "statistics" shares a root with "state." It began in the 18th century as the numerical bookkeeping of nations: population, taxes, soldiers, harvests. The mathematics of describing variation grew out of the practical needs of governments counting their citizens, long before it became a tool of science.

EXT Your calculator returns r without showing its working. The formula behind it is

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}},$$

and it is worth reading rather than memorising.

Take one term of the numerator, $(x_i - \bar{x})(y_i - \bar{y})$. It asks where a single point sits relative to *both* means. Above average in both, or below average in both, and the product is positive; above in one and below in the other, and it is negative. So each point votes, and the numerator is the total vote.

That total depends on the units, and the denominator divides them out, so r is a pure number. It also forces $-1 \leq r \leq 1$: the numerator can never exceed the denominator, a fact called the Cauchy–Schwarz inequality, and equality holds exactly when every point lies on one straight line. That is why $r = \pm 1$ means perfect linearity, and nothing else does.

End-of-Chapter Problems — Chapter 11: Statistics

1. The times, in minutes, taken by 40 students to complete a puzzle are summarised below.

Time (min)	$0 \leq t < 10$	$10 \leq t < 20$	$20 \leq t < 30$	$30 \leq t < 40$	$40 \leq t < 50$
Frequency	4	11	15	8	2

- (a) Estimate the mean time. [3]
 (b) Write down the modal class. [1]
 (c) State the class containing the median. [2]

2. A data set is 12, 15, 18, 19, 21, 22, 24, 25, 27, 28, 30, 46.

- (a) Find the five-number summary. [3]
 (b) Determine whether 46 is an outlier, showing your working. [3]
 (c) Describe the shape of the distribution. [1]

3. GDC The table shows the number of hours x eight students revised and their test score y .

x	2	3	5	6	8	9	11	12
y	10	14	15	20	24	27	30	34

- (a) Find Pearson's correlation coefficient and interpret it. [3]
 (b) Find the equation of the regression line of y on x . [2]
 (c) Interpret, in context, the gradient and the y -intercept of your line. [2]
 (d) Use your line to estimate the score of a student who revised for 7 hours. [2]
 (e) Explain whether your line should be used to estimate the score for 25 hours. [2]

4. A set of data has mean 45 and standard deviation 8.

- (a) Each value is increased by 12. Write down the new mean and standard deviation. [2]
 (b) Instead, each value is multiplied by 1.5. Write down the new mean and standard deviation. [2]
 (c) Explain why adding a constant changes the mean but not the standard deviation. [2]

5. A region has populations of 4000, 10 000, and 6000 in three towns A, B, and C. A stratified sample of 100 households is required.

- (a) Find the number of households to sample from each town. [3]
 (b) State one advantage of stratified sampling over simple random sampling here. [2]

6. The cumulative frequencies of a data set of 60 values are given.

Value $\leq$	10	20	30	40	50
Cumulative frequency	4	16	34	52	60

- (a) Draw a cumulative frequency curve. [3]
 (b) Use it to estimate the median and the interquartile range. [4]
 (c) Estimate the 70th percentile. [2]

7. In a test, 20 students in class A had a mean score of 72, and 30 students in class B had a mean score of 65. Find the mean score of all 50 students combined. [3]

8. A frequency distribution of the values 1, 2, 3, 4 has frequencies 5, 8, k , 4. The mean is 2.5. Find the value of k . [4]

9. Two machines fill bottles. A sample from each gives the same mean volume of 500 mL, but machine X has standard deviation 2 mL and machine Y has standard deviation 9 mL.

- (a) Explain what this tells you about the two machines. [2]
 (b) State which machine is more reliable, with a reason. [2]

10.  The table gives the area x (in m^2) and monthly rent y (in \$) of five apartments.

x	10	20	30	40	50
y	14	20	25	33	38

- (a) Find the regression line of y on x . [2]
 (b) Use it to estimate the rent of a 35 m^2 apartment. [2]
 (c) Find the regression line of x on y , and use it to estimate the area of an apartment renting for \$30. [3]

11. A data set is 4, 6, 7, 8, 9, 10, 55.

- (a) Find the mean and the median. [3]
 (b) The value 55 is removed. Find the new mean and median. [2]
 (c) Comment on which measure of centre is more affected by the outlier, and why. [2]

12. A student's final grade is 40% coursework and 60% final exam. The student scores 70 on coursework and 85 on the exam. Find the final grade. [2]

13. The masses, in grams, of seven eggs are 52, 55, 58, 60, 61, 63, 90.

- (a) Find Q_1 , Q_3 , and the interquartile range. [3]
 (b) Show that 90 is an outlier. [2]
 (c) State, with a reason, whether the mean or the median better represents a typical egg. [2]

14. A regression line of y on x passes through the mean point $(6, 19)$ and has gradient -2 .

- (a) Find the equation of the regression line. [2]
 (b) Use it to predict y when $x = 4$. [2]

15. The five-number summaries of test scores in two classes are given below.

	Min	Q_1	Median	Q_3	Max
Class A	32	45	55	62	90
Class B	42	50	58	66	74

- (a) Write down the interquartile range for Class A. [1]
 (b) Show that the score of 90 in Class A is an outlier. [2]
 (c) Compare the two distributions, referring to centre and spread. [2]
 (d) State, with a reason, whether the scores in Class B may be normally distributed. [2]

16. GDC The waiting times, in minutes, of a group of customers are summarised below. The estimated mean waiting time is 20 minutes.

Time (min)	$0 \leq t < 10$	$10 \leq t < 20$	$20 \leq t < 30$	$30 \leq t < 40$
Frequency	6	k	12	8

- (a) Show that $k = 18$. [3]
 (b) Write down the modal class. [1]
 (c) State the class containing the median. [2]
 (d) Use your GDC to estimate the standard deviation. [2]

17. A set of examination marks x has mean 62 and standard deviation 10. The marks are scaled linearly to $y = ax + b$, $a > 0$, so that the scaled marks have mean 70 and standard deviation 8.

- (a) Find the values of a and b . [4]
 (b) A student's original mark was 75. Find their scaled mark. [1]
 (c) The median of the original marks was 61. Write down the median of the scaled marks. [1]

18. GDC The table shows the maximum temperature x ($^{\circ}\text{C}$) and the number of ice creams sold y on six days.

x ($^{\circ}\text{C}$)	18	20	23	25	27	30
y	110	125	150	160	175	195

- (a) Find Pearson's correlation coefficient r and comment on it. [2]
 (b) Find the equation of the regression line of y on x . [2]
 (c) Interpret, in context, the gradient of your line. [1]

- (d) Estimate the number of ice creams sold on a day with maximum temperature 24°C . [2]
 (e) Explain why the line should not be used when the maximum temperature is 40°C . [1]

19. For a bivariate data set, the regression line of y on x is $y = 2x + 1$ and the regression line of x on y is $x = 0.4y + 1$.

- (a) Find $\bar{x}$ and $\bar{y}$. [4]
 (b) Estimate the value of x when $y = 20$. [2]
 (c) Explain why the regression line of y on x should not be used in part (b). [1]

20. A supermarket wants to survey its customers.

- (a) State the sampling method used when every 20th customer entering the store is selected. [1]
 (b) State the sampling method used when customers are grouped by age band and sampled in proportion to the size of each band. [1]
 (c) State the sampling method used when the first 30 customers on Monday morning are selected. [1]
 (d) Explain why the method in (c) is likely to produce a biased sample. [2]
 (e) A school of 320 juniors and 480 seniors is to be surveyed with a stratified sample of 50. Find the number of juniors and seniors in the sample. [2]

21. A group of n students has a mean mark of 68. When 5 students with mean mark 60 join the group, the overall mean falls to 66. Find n . [4]

22.

- (a) Match each value of r to one description: $r = -0.92$, $r = -0.35$, $r = 0.08$, $r = 0.85$.
 Descriptions: strong positive, weak negative, strong negative, no linear correlation. [2]
 (b) A study finds a strong positive correlation between the hours children spend watching television and their body mass. Explain why this does not show that television causes weight gain, and suggest a possible third variable that could explain both. [2]
 (c) Heights are recorded in centimetres and r is calculated. The heights are converted to metres. State, with a reason, the effect on the value of r . [2]

23. The table gives the monthly mean air temperature in Tokyo, in $^{\circ}\text{C}$, taken from the Japan Meteorological Agency climatological normals for 1991–2020.

Month	J	F	M	A	M	J	J	A	S	O	N	D
Temp ($^{\circ}\text{C}$)	5.4	6.1	9.4	14.3	18.8	21.9	25.7	26.9	23.3	18.0	12.5	7.7

- (a) Find the five-number summary of the twelve values. [4]

- (b) Show that the data contain no outliers. [3]
- (c) The JMA publishes an annual normal of 15.8°C . Calculate the mean of the twelve monthly values and compare it with the published figure. [3]
- (d) Describe the shape of the distribution, and explain why this box plot tells you very little about the temperature on any particular day in Tokyo. [3]

24. A data set of n values has mean $\bar{x}$. A new value equal to $\bar{x}$ is added to the set.

- (a) Show that the mean is unchanged. [3]
- (b) Show that the standard deviation decreases (or stays the same). State the condition for equality. [4]

25. Five positive integers have a mean of 8, a median of 7, and a mode of 4 (occurring exactly twice). Find all possible sets of five integers. [6]

26. The regression line of y on x for a data set is $y = 0.8x + 3$, and the regression line of x on y is $x = 1.05y - 2$.

- (a) Both lines pass through $(\bar{x}, \bar{y})$. Find $\bar{x}$ and $\bar{y}$. [4]
- (b) It can be shown that the product of the two gradients equals r^2 . Find r . [3]

27. A set of n data values has mean $\bar{x}$ and standard deviation σ . Prove algebraically that adding a constant c to every value leaves σ unchanged, starting from the definition $\sigma^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$. [5]

Challenge Questions

28. **How much can one value move a summary?** Consider the data set

3, 5, 7, 9, 11.

- (a) Write down the mean and the median. [2]
- (b) The largest value is replaced by k , where $k \geq 9$. Find the mean in terms of k , and write down the median. [4]
- (c) Find the value of k for which the mean is 20. [2]
- (d) Explain why the median is 7 for every $k \geq 9$. [3]
- (e) A data set has n values. Explain why changing a single value can move the mean by any amount, however large, but can move the median by at most one place in the ordered list. Hence explain what is meant by calling the median *resistant* and the mean *not resistant*. [5]
- (f) State, in terms of n , the smallest number of values that must be changed to make the median as large as you wish. Justify your answer for $n = 5$. [3]

29. Combining two groups. Group A has n_1 values with mean $\bar{x}_1$ and variance σ_1^2 . Group B has n_2 values with mean $\bar{x}_2$ and variance σ_2^2 . The two groups are combined into one set of $n_1 + n_2$ values.

- (a) Show that the combined mean is $\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2}$. [3]
 (b) Group A has 12 values with mean 50, and Group B has 8 values with mean 60. Find the combined mean. [2]
 (c) Using $\sum x^2 = n(\sigma^2 + \bar{x}^2)$ for each group, show that the combined variance is

$$\sigma^2 = \frac{n_1(\sigma_1^2 + \bar{x}_1^2) + n_2(\sigma_2^2 + \bar{x}_2^2)}{n_1 + n_2} - \bar{x}^2.$$

- [5]
 (d) Group A has standard deviation 4 and Group B has standard deviation 3. Find the standard deviation of the combined set. [4]
 (e) Show that $\sigma^2 \geq \frac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2}$, and state the condition for equality. Interpret this result. [4]

30. Putting data on a common scale. A data set $x_1, x_2, \dots, x_n$ has mean $\bar{x}$ and standard deviation σ , with $\sigma \neq 0$. Each value is replaced by its *standardised value*

$$z_i = \frac{x_i - \bar{x}}{\sigma}.$$

- (a) Show that the mean of the z_i is 0. [4]
 (b) Show that the standard deviation of the z_i is 1. [4]
 (c) The data set 4, 8, 10, 12, 16 has mean 10 and standard deviation 4. Write down the five standardised values, and verify your answers to parts (a) and (b). [4]
 (d) Student P scores 78 on a test with mean 70 and standard deviation 5. Student Q scores 84 on a different test with mean 75 and standard deviation 8. Determine which student performed better relative to their own group. [4]
 (e) Explain why standardised values allow marks from two different tests to be compared, when the raw marks cannot be. [2]

